

AI-Driven Prediction of Indian Criminal Case Outcomes

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Abstract—Indian courts have too many cases and not enough resources or judges, causing delays and undermining trust in the legal system. In addition, Indian law is quite complex, requiring law students and legal professionals to study and understand it thoroughly for better practice. To address these challenges, we have developed an AI-powered legal assistant designed to enhance the efficiency and effectiveness of legal processes. The three main features of our proposed approach are judgment prediction, case summarization, and IPC section and punishment prediction. The judgment prediction uses AI & ML to analyze case facts and give the probability of each party winning. The case summarization compacts the long legal documents into concise texts, highlighting key legal terms and facts. The IPC section and punishment prediction identifies applicable sections of the Indian Penal Code and predicts. These features enhance legal workflows, reduce manual document analysis and improve legal decision-making. Our proposed legal AI system is tested by deploying on Streamlite, which provides interactive interface that can be used on any device with internet access. The performance of the proposed AI model is analysed and the general accuracy of the classification model is obtained as 97%. The performance can be further improved using a more comprehensive and diverse dataset. The consistency and reliability of the model are validated by computing macro- and weighted averages, affirming its robust performance across all IPC categories. In addition, the precision, recall, and F1 score values indicates the accurate identification of winning parties.

Index Terms—Legal Assistant, AI, Machine Learning, Judgment Prediction, Case Summarization, Indian Penal Code, Punishment Prediction, Legal Efficiency, Legal Decision Making, Judicial System.

I. INTRODUCTION

AI is increasingly integrated into daily life across sectors, including healthcare, finance, and now the legal domain. In the Indian judiciary, AI will help the judiciary of India deal with issues such as case backlogs and resource constraints, improving efficiency and public trust. In the context of the legal domain, AI has the potential to address some of the most pressing challenges faced by Indian courts, such as the long backlog of cases and the shortage of judges and resources. The Indian judiciary system is currently facing problems with significant delays in cases, which not only degrades the efficiency of the legal process but also decreases public trust. In addition, the complexities of Indian law require deep understanding and study by

lawyers and students alike [1]. These challenges led us to this innovative solution that can make legal workflows easier, reduce manual burdens, and improve decision-making accuracy. The major features of the proposed solution are the Simplification of Legal Operations, Reducing Manual Duties, and Decision Making with Better Accuracy. The Indian Penal Code(IPC) is the criminal code in India, which is a legal framework maintaining law and order in India. This addresses a wide range of crimes, including theft, assault, murder, and fraud, providing a legal framework to maintain law and order. The IPC ensures justice by defining criminal acts and establishing consistent legal procedures and penalties, playing a crucial role in the Indian judicial system.

In this paper, we present a model that can generate the summarization of the case using simple terms so that people other than lawyers can understand the case documents. Imagine a case involving a property issue between two parties. By entering the case facts, along with the names of the petitioner and respondent, our proposed model can predict the probability of each party winning. Like petitioner has a 70% chance of winning based on historical data and case precedents. That is the prediction feature of the judgment. Suppose a lawyer is handling a case with a large file. Our model can quickly summarize documents, highlighting key legal terms and essential case facts. This summary allows the lawyer or generalist to grasp the critical points without referring to hundreds of pages, thus saving valuable time and effort. That is the Case Summarization feature. Consider a criminal case where the accused is charged with theft. By analyzing the details of the case, our model can identify the relevant sections of the Indian Penal Code (IPC) and predict the potential punishments. For instance, it might indicate that the accused can be held accountable under IPC Section 379 and suggest the corresponding punishment. That is the IPC Section and Punishment Prediction feature. The proposed model is trained using multiple types of IPC case documents with their judgments. Fig.1 presents the workflow of the proposed work.

The paper is organized as follows. Section 2 reviews the related work, and Section 3 describes the features of our model. The results are presented in Section 4. In Section 5,

possible improvements for the model and the conclusion are presented.

II. RELATED WORK

The prediction of legal judgment has attracted considerable attention in the fields of legal informatics and artificial intelligence due to the sheer volume of legal cases and the necessity of efficient legal decision-making processes. Intisar Almuslim and Diana Inkpen [2] proposed a system that addresses the problem of predicting the outcomes of court cases using natural language processing (NLP) and machine learning (ML). Legal professionals often struggle with large volumes of case law, so they set out to create a smart system. J. Sulea et al. [3] introduced a method that uses NLP and ML techniques to predict the results of cases in the French Supreme Court. Li et al. [4] explored a method that focuses on the citation network of legal cases to predict court decisions. G. Sukanya et al. [5] have used a hybrid self-improved Recurrent Neural Network (RNN) that is trained in 10 types of real-time criminal cases. It is a domain-specific word embedding model.

Other works address the ethical and interoperability concerns associated with the use of AI in the prediction of legal judgments. T. Bench-Capon and F. Coenen [6] highlighted the need for explainable AI systems in the legal domain. However, J. Niklaus et al. [7] have presented a multilingual judgement prediction in various languages and have trained using the BART model. They have achieved 68-70% Macro-F1 score in German and French languages.

By integrating advanced NLP techniques with domain-specific features and explainable AI (XAI) methods, we aim to develop a robust system. Our proposed system goes beyond simply predicting the outcomes of the cases. It also identifies applicable sections within the law and appropriate punishments.

III. IMPLEMENTATION OF THE PROPOSED AI LEGAL SYSTEM

Fig.1 shows the different tasks to be performed to implement the proposed AI legal system. The implementation of each task is explained in the following subsections.

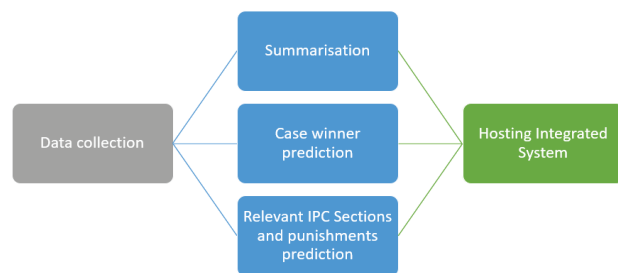


Fig. 1. Workflow of the Proposed Legal Case Analysis System

A. Legal Text Summarization

The data set containing Indian court judgment-summary pairs serves as the foundational source for the summarization task of our proposed work and is instrumental in training the BART model [8]. With 6919 entries, each pairing a judgment with a concise summary in legal terminology.

The Bidirectional and Auto-Regressive Transformers (BART) model is selected for our summarization task due to its effectiveness in generating coherent and contextually relevant summaries from lengthy legal texts. Facebook AI [9] created BART, a cutting-edge transformer-based architecture that performs exceptionally well in a variety of natural language processing applications. This bidirectionality allows BART to understand the full context of a legal judgment, which is crucial for generating precise summaries. Furthermore, BART employs an auto-regressive decoding mechanism, where it generates one token at a time while conditioning on previously generated tokens. This approach ensures that the generated summary flows smoothly and maintains coherence with the original text. BART is also pre-trained on large collections of text data, including diverse sources such as news articles, books, and websites. Its conditional generation capability allows us to control the length and quality of the generated summaries, ensuring that they meet our requirements for concise summaries of legal judgments. The BART model is trained on the aforementioned data set and utilizes a beam search with 4 beams to generate a summary with a maximum length of 150 tokens. In addition, early stopping is implemented to improve efficiency during the summarization process. With this approach, the model accommodates input text of up to 1024 tokens in length, allowing us to obtain a succinct summary comprising approximately 150 words.

B. Prediction of Applicable IPC Sections and Corresponding Punishments

Our proposed approach shown in Fig.2 involves developing a model to predict sections of the Indian Penal Code (IPC) and potential punishments based on the facts of criminal cases. The proposed model is developed by employing a combination of Natural Language Processing (NLP) techniques and machine learning algorithms to achieve this goal. Our proposed approach involves developing a model to predict sections of the Indian Penal Code (IPC) and potential punishments based on the facts of criminal cases. The proposed model is developed by employing a combination of Natural Language Processing (NLP) techniques and machine learning algorithms to achieve this goal. Initially, we read and processed our dataset, which consists of various criminal cases, each associated with an offense description, the corresponding IPC section, and the related punishment. We handled potential errors in the dataset by implementing a mechanism to skip any invalid lines while reading the file. To prepare the text data for modeling, several pre-processing steps are employed, including tokenization, removal of stop

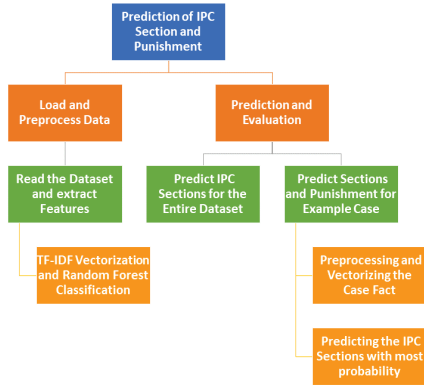


Fig. 2. Relevant IPC section and punishment Prediction

words, and lemmatization. Tokenization splits the text into individual words, stop words removal eliminates common but uninformative words, and lemmatization reduces words to their base forms. This preprocessing ensured that the text data are in a suitable format for feature extraction. For feature extraction, the term frequency-inverse document frequency vectorizer (TF-IDF) is used. The TF-IDF vectorizer transforms the text data into numerical vectors that reflect the importance of each word in the documents relative to the entire dataset.

The Random Forest classifier [10] is selected for the prediction task due to its robustness and effectiveness in handling complex data. The classifier can be described as follows:

$$\hat{y} = \operatorname{argmax} \sum_{t=1}^T I(h_t(x) = y) \quad (1)$$

where $\hat{y}$ is the predicted class, T is the total number of trees, h_t is the decision tree t , and I is an indicator function that outputs 1 when $h_t(x) = y$ and 0 otherwise.

Each decision tree h_t in the forest is built using a bootstrap sample of training data and a random subset of features. This process introduces diversity among trees and helps prevent overfitting.

Furthermore, we implemented a function to predict the IPC sections and potential punishments for new case facts. This function preprocesses the input text, transforms it using the TF-IDF vectorizer, and utilizes the trained Random Forest model to predict the IPC sections. The results of our evaluation showed that the proposed model could effectively predict IPC sections with a reasonable degree of accuracy. However, accuracy can be further improved by obtaining a more comprehensive and diverse dataset.

C. Estimation of Case Winner Probability

TF-IDF is probably the most prominent NLP technique for measuring the importance of words. It consists of two elements: term frequency and inverse document frequency.

TF-IDF highlights terms that are frequent in a document but not throughout the corpus, hence capturing major terms while discarding common ones. TF-IDF is used to make a probability prediction concerning winners in legal matters.

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

$$IDF(t, D) = \log \left(\frac{\text{Total number of documents in corpus } D}{\text{Number of documents containing term } t} \right)$$

$$TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

1) *Training of Model:* For text classification applied to legal documents, each step does:

- **Anonymization:** It's a function to replace specific names or entities in the text with generic placeholders to anonymize the information while preserving its structure and meaning.
- **Text Vectorization:** We created a function that is used to convert anonymized text data into numerical feature vectors using the TF-IDF method. This transforms text data into a suitable format for ML algorithms by representing each document as a vector of TF-IDF values.
- **Model Training:** - The function is called to train a classification model using the TF-IDF vectors and the corresponding labels. This function is trained on the training data, with performance metrics such as loss and accuracy being recorded for evaluation.
- **Evaluation:** - Performance metrics such as loss and accuracy are likely recorded during training are done to monitor the model's performance on both the training and validation sets.

2) *Custom TF-IDF Vectorization for Case Data:* In general, this code snippet represents a typical workflow for training a text classification [3] model, including data preprocessing (anonymization), feature engineering (TF-IDF vectorization), model training, and performance evaluation. In our proposed approach, we leverage a TF-IDF pre-trained model to analyze data from legal cases and predict the probability that each party will win. Initially, we pre-process the data set by tokenizing the text and anonymizing sensitive information to ensure privacy and eliminate biases. This preprocessing step is crucial to prepare the data for modeling. With the preprocessed data set in hand, a customized TF-IDF vectorizer generated to our specific data set.

Using the TF-IDF pretrained model, we predict the probabilities that each party prevailing in the legal dispute. The model outputs these probabilities as a numpy array, offering insights into the potential outcomes for both parties involved.

3) *Use of XAI:* Explainable Artificial Intelligence (XAI) aims to make complex machine learning models more understandable to humans. LIME (Local Interpretable Model-agnostic Explanations) is a popular XAI technique that

provides explanations for individual predictions by approximating the model's behavior locally around them. In this work, LIME is integrated as part of XAI to enhance transparency and interpretability in predicting legal judgments. By highlighting the most influential words in a case's facts, LIME helps users understand why the model made a specific prediction. This approach fosters trust and accountability by providing insight into the legal decision-making process of the machine learning model, which is crucial in legal contexts where transparency is of the utmost importance.

D. Interface Overview for Legal AI System

The proposed legal AI system, deployed on Streamlit, provides an accessible and interactive interface that can be used on any device with Internet access, including phones, laptops, and tablets. The user interaction with the system can be summarized as follows.

1) *Accessing the Interface:* Users can access the Legal AI system through a Streamlit link, which opens the web application. The interface is user-friendly, allowing for seamless navigation and interaction.

2) *Input Methods:* Upon accessing the system, users are presented with the option to enter case facts through two methods:

- Upload a Text File: Users can upload a.txt file containing the case details. The system processes the file to extract necessary information such as the petitioner, respondent, charges, and facts of the case.
- Type in text form: Users can manually enter the details into provided text fields:
 - *Petitioner:* The name of the petitioner.
 - *Respondent:* The name of the respondent.
 - *Charges:* The charges to be filed.
 - *Case Facts:* A detailed description of the facts of the case.

3) *Prediction Process:* Once the user provides the necessary input, they can submit the form to trigger the prediction process. The system performs the following steps:

- Case Facts Preprocessing: The entered case facts are preprocessed to remove stop words and anonymize the names of the petitioner and respondent.
- Model Predictions: The system uses a pre-trained machine learning model to predict the likelihood of each party winning the case. The predictions are displayed as percentages for both the petitioner and the respondent.
- Case Summary: A summarized version of the case facts is generated and presented to the user for easy reference.
- Highlighted Words: The interface highlights the most influential words in the case facts that contributed to the model's decision, providing transparency in the prediction process.
- Section and Punishment Prediction: The system predicts relevant IPC sections and the corresponding punishments based on the charges and case facts provided.

This information is clearly displayed, indicating the applicable sections and potential consequences.

4) *Display of Results:* The results are presented in a structured format, making it easy for users to interpret the predictions.

- Winning Probabilities: The probabilities of the petitioner and respondent winning are displayed side by side.
- Winning Party: The system declares the party with the higher probability as the predicted winner.
- Highlighted Words: The most significant words influencing the model's decision are shown, color-coded based on their impact.
- IPC Sections and Punishments: The relevant IPC sections and associated punishments are listed, providing a comprehensive overview of the legal implications.

The system can predict the likely outcome and identify applicable sections and sentences by allowing users to input the facts and charges of the case. Y. Le et al. [11] worked on a framework for ranking the prediction.

IV. RESULTS

The analysis of the results generated by the proposed AI legal model presents the performance metrics for the classification of five categories of Indian Penal Code (IPC), as shown in Table 1, providing information on precision, recall, and F1 score for each IPC section, along with their corresponding support values. The table underscores the effectiveness of the classification model, revealing strong performance across the board. The general accuracy of the classification model is high at 97%, based on a total support count of 4034 instances. In addition, both the macro- and weighted averages validate the consistency and reliability of the model, affirming its robust performance across all IPC categories.

TABLE I
PERFORMANCE METRICS FOR IPC SECTION AND PUNISHMENT PREDICTION

IPC	Precision	Recall	F1-score	Support
IPC_208	0.75	0.60	0.67	10
IPC_209	0.80	0.89	0.84	9
IPC_210	1.00	0.92	0.96	12
IPC_211	0.86	1.00	0.92	6
IPC_212	1.00	0.80	0.89	5
Accuracy	0.97			4034
Macro avg	0.97	0.97	0.97	4034
Weighted avg	0.97	0.97	0.97	4034

As we can see in Table II, the probability prediction of the winners using the TF-IDF model show strong performance, with precision, recall, and F1 score values indicating accurate identification of the winning parties.

Fig.3 & Fig.4 presents the screenshot of the output generated by the proposed AI legal system for the case details provided to it as input.

TABLE II
PERFORMANCE METRICS FOR CASE WINNER PROBABILITY
PREDICTION

	Precision	Recall	F1-Score	Support
Class 0	0.99	0.94	0.96	423
Class 1	0.92	0.98	0.95	270
Accuracy			0.96	693
Macro Avg	0.95	0.96	0.96	693
Weighted Avg	0.96	0.96	0.96	693

Back Legal Judgement Prediction

Case Summary

Dr Priyanka Reddy was ambushed by four men in Hyderabad. They sexually assaulted her, poured whiskey mouth silence screams, eventually smothered death. Police arrested four men suspects based CCTV footage evidence victim's mobile phone. The accused identified Chintakunta Chennakeshavulu, Jolu Shiva, Jollu Naveen, Mohammed Arif.

Winning Party

priyanka reddy

Percentage of petitioner winning

99.93002401315607%

Percentage of respondent winning

0.0699759868439287%

Highlighted Text

On evening 27th November 2019, Priyanka Reddy parked scooter near toll plaza Shamshabad **took** taxi visit dermatologist's office Hyderabad. While away, **four** men noticed scooter hatched plan assault her. When Priyanka returned around 9:15 p.m., found scooter's tire deflated. **The** men pretended help offered assistance. However, instead ambushed her, pushed nearby bushes, turned

Fig. 3. Example Criminal Case Output

A. Example

V. CONCLUSIONS

The vastness and complexity of Indian laws posed significant challenges during our project development. With no suitable dataset available for Indian Crime Cases, we initially considered focusing on Consumer Laws but found dataset selection too complex due to variations across products and regions. Consequently, we narrowed our focus to criminal law for better manageability. The proposed model is developed by employing BART for summarizing and Random-Forest classifier for predicting IPC sections and TF-IDF approach for winner predictions. Since there is no suitable data set available, we created our own on the basis of IPC sections. We also took advantage of Streamlit's free version to host our model without additional costs. While our approach for IPC section and corresponding punishment prediction performs well for shorter texts using our curated dataset from IPC official documents, for longer

case documents, leveraging language models like LLMs could be beneficial for analysis of verdicts. Despite the challenges posed by the complexity of legal systems and the scarcity of suitable datasets, our efforts have resulted in a promising framework. We can streamline legal procedures,

victim, tragically, partially burnt **body** discovered bridge approximately 25 kilometres away location last seen. Priyanka Reddy's charred **body** discovered Chatanpally Bridge Shadnagar 28th November 2019, day incident. Following investigation, **police** **arrested** **four** men suspects based CCTV footage evidence victim's mobile phone. **The** **accused** identified Chintakunta Chennakeshavulu, Jolu Shiva, Jollu Naveen, Mohammed Arif. **The** **accused** **confessed** crime, taken judicial custody. **The** Chief Minister Telangana ordered formation fast-track court expedite trial. **The** **police** **arrested** **four** individuals later **confessed** involvement crime. However, case **took** dramatic turn **four** convicts killed **police** encounter, sparked wide range reactions debates India regarding encounters extrajudicial killings.

Applicable Sections

IPC_376, IPC_307, IPC_204, IPC_201, IPC_195A

Punishments

Imprisonment for life or death, 10 Years + Fine, Up to 2 Years Imprisonment, Fine, or Both, 7 Years Imprisonment + Fine, Imprisonment up to 7 Years and Fine

Fig. 4. Continuation of Output

leading to a more efficient and equitable distribution of justice by leveraging AI capabilities.

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